

02 - NoSQL and the Document Database

CSCI 112 - Contemporary Databases

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Learning Objectives

- Understand the theory behind NoSQL databases
- Understand the NoSQL BASE model
- Contrast SQL and NoSQL data models

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Scenario for the Day



Context Setting

- *The Coco Group* was happy with the initial *Kahit Saan* engagement. Now, they wanted to add 2 more companies from their portfolio to the digital transformation initiative. They have actually named the initiative to make it official: Olympus 2030.
- The 2 companies are Rural Savings, a rural bank in a town with approximately 100,000 population, and Fitness Last, a commercial gym with branches all over the country.
- Rural Savings would like to start their digital transformation with PRIME, their core banking system for retail credit card accounts.
- Meanwhile, Fitness Last would just like a simple app to test the waters of this initiative. The app should be able to show members of the gym how many people are in a particular gym location at a given time.

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CAP Theorem



CAP Theorem

CONSISTENCY

Every read receives the most recent write, all nodes see the same data at the same time.

AVAILABILITY

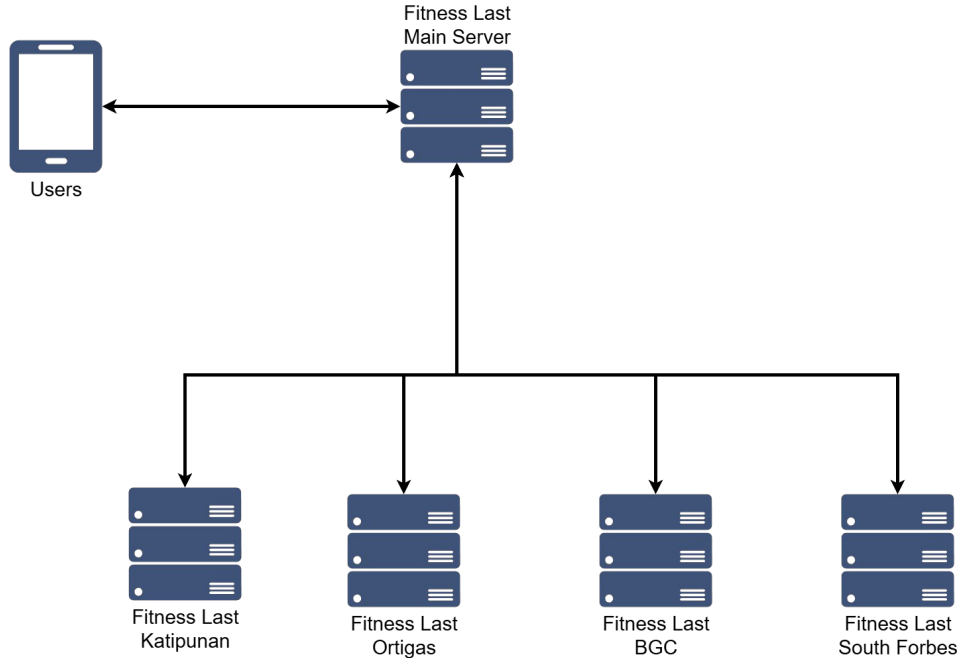
The system operates and responds to requests even if some nodes are down which may return stale data.

PARTITION TOLERANCE

The system continues to function despite network partitions or downtimes.



CAP Theorem (AP Configuration)



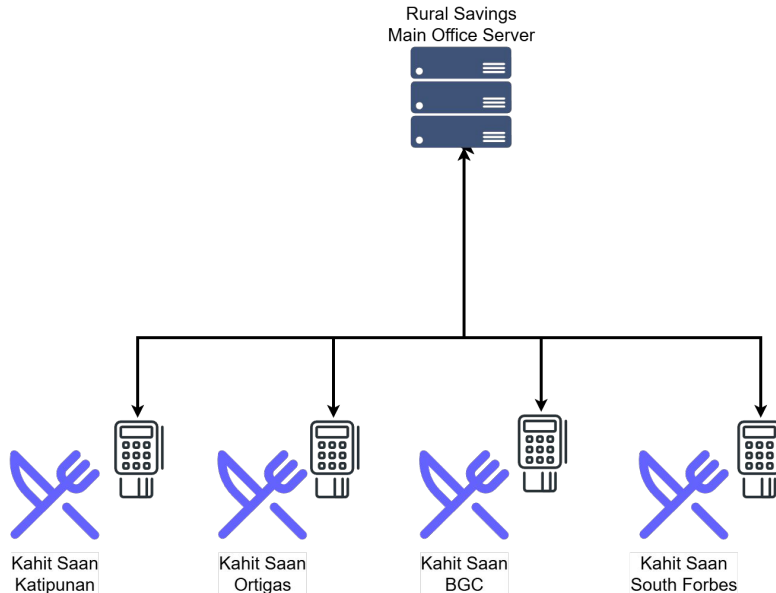
Users access data that is directly stored in the main (central) server.

What happens when a user from the Katipunan branch wants to see how many members are currently onsite and the network connection to the Katipunan branch is unavailable?

The main server accesses data from the branch servers whenever a network connection is available. If none, then it uses the last stored data to fulfill requests from the users.



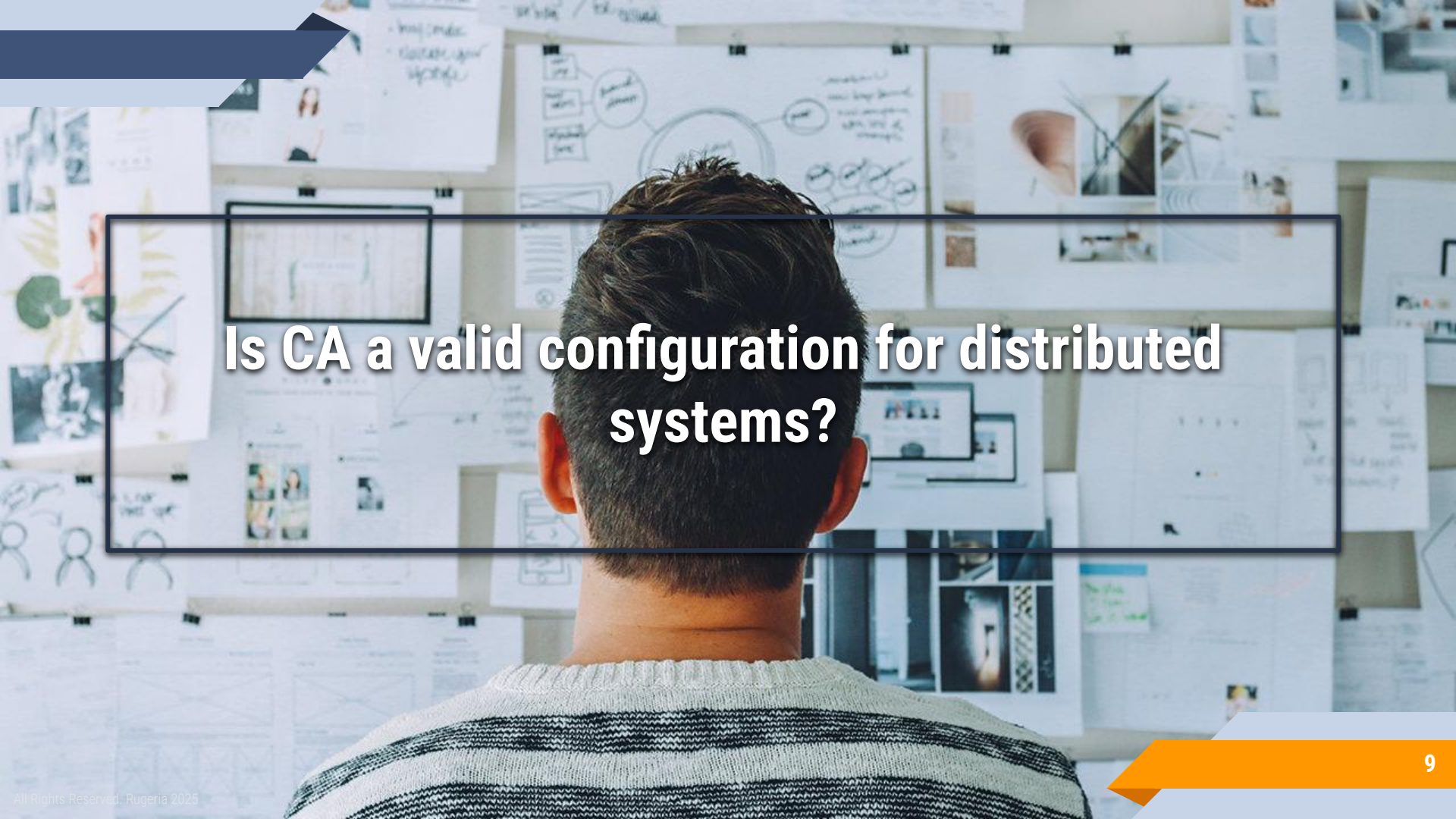
CAP Theorem (CP Configuration)



Credit Card Terminals situated in different *Kahit Saan* branches directly connect to the main office server to retrieve the most recent data on balances.

What happens when a user from the Katipunan branch transacts using their credit card and a network connection to the main office is unavailable?

When there is a network partition (unavailability), the terminals should return an error.

A person with short brown hair, seen from the back, is looking at a wall covered in various papers, diagrams, and photos. The papers contain handwritten notes, sketches, and printed images. A large, semi-transparent white box with a dark border is centered over the person's head and the wall, containing the text "Is CA a valid configuration for distributed systems?".

Is CA a valid configuration for distributed systems?

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Database Transaction Behaviors



ACID

ACID behavior ensures database transactions are reliable (CP configuration)

- **Atomic** - db transactions are either fully executed, or rolled back when the operation fails.
- **Consistent** - db transactions maintain data integrity (data types, constraints, referential integrity, business rules, etc.)
- **Isolated** - db transactions do not interfere with each other
- **Durable** - once a db transaction is committed, it is permanent



Handles large-scale distributed systems (AP Configuration)

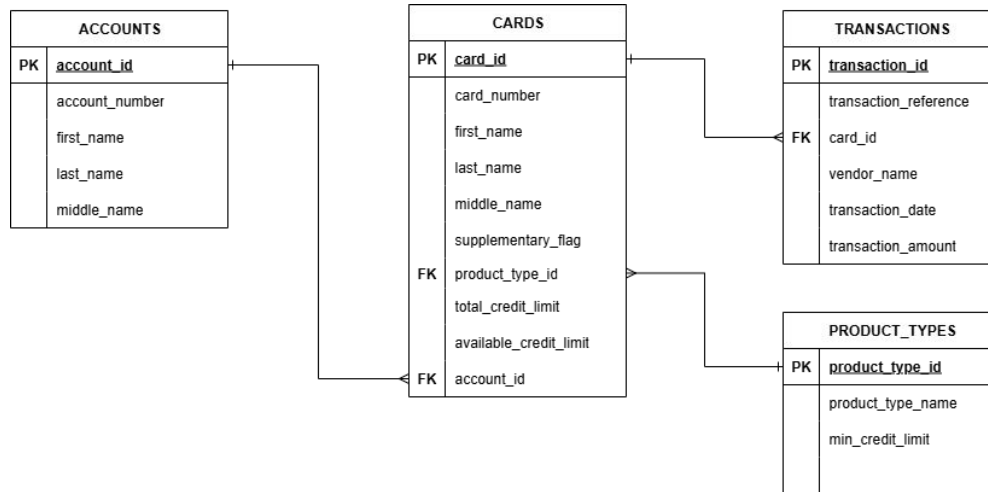
- **Basic Availability** - the db guarantees availability, even during downtimes
- **Soft State** - the db's state may change over time even without new inputs/transactions
- **Eventual Consistency** - the db ensures that eventually, all replicas will have the same, consistent data

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SQL vs NoSQL Schemas



Sample SQL Schema



The ERD on the left shows a sample SQL model that could represent the database that Rural Saving's uses for PRIME (Credit Card Core Banking System)

Can you describe the relationship between the entities?

If you wanted to create a report on card transactions per account and card, which tables and fields would you include?



Sample SQL Schema

transaction_id	transaction_reference	vendor_name	transaction_date	transaction_amount	card_number	product_type_name	account_number	last_name
1	RS0000000001	Kahit Saan Katipunan	2025-07-01	580.00	0000 0000 0000 2987	Platinum	00001	Spender
2	RS0000000002	Fitness Last South Forbes	2025-07-02	3500.00	0000 0000 0000 2987	Platinum	00001	Spender
3	RS0000000003	Kahit Saan BGC	2025-07-03	1490.50	0000 0000 0000 1237	Standard	00001	Spender
4	RS0000000004	Kahit Saan BGC	2025-07-04	2568.90	0000 0000 0000 0928	Gold	00001	Spender



Sample NoSQL Schema

Rural Savings PRIME Database

```
{  
  transaction_reference: "RS0000000001",  
  vendor_name: "Kahit Saan Katipunan",  
  transaction_date: "2025-07-01",  
  transaction_amount: 580.00  
  card_number: "0000 0000 0000 2987",  
  product_type_name: "Platinum"  
  account_number: "00001",  
  last_name: "Spender"  
}
```

If you wanted to create a report on card transactions per account and card, which tables and fields would you include?

– This is an access pattern (more on this in the next module)

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SQL vs NoSQL



SQL vs NoSQL Comparison

Category	SQL	NoSQL
Structure	Multiple tables, each containing columns and rows	Single Collection, each having documents with keys and values
Schema	Structured and Strict	Semi-structured and Flexible
Querying	Structured Query Language (SQL)	Depends on the database service
Scaling	Vertical	Horizontal
Transactions	ACID	BASE
Performance	Complex Queries with Joins and Aggregations may affect performance	Optimized for fast read/write operations

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Document Database and MongoDB



Document Database

- Stores records as “documents”
- Documents are stored in a Collection
- Documents are in the form of key-value pairs



Structure

What's inside a Document Database?





Database – Logical Container of Collections

- Contains multiple collections
- Organizes collections into a logical grouping
- Implicitly created –created when a collection is created within the database



Collection – Fundamental Organizational Unit

- Equivalent of tables in a SQL database
- Stores multiple pieces of data (documents)
- Implicitly/Explicitly created –created at the first transaction and does not have to be created before that unlike SQL tables



Document – A Single Record in a Document DB

- Equivalent of rows in a SQL database
- Contains a JSON schema
- Explicitly created

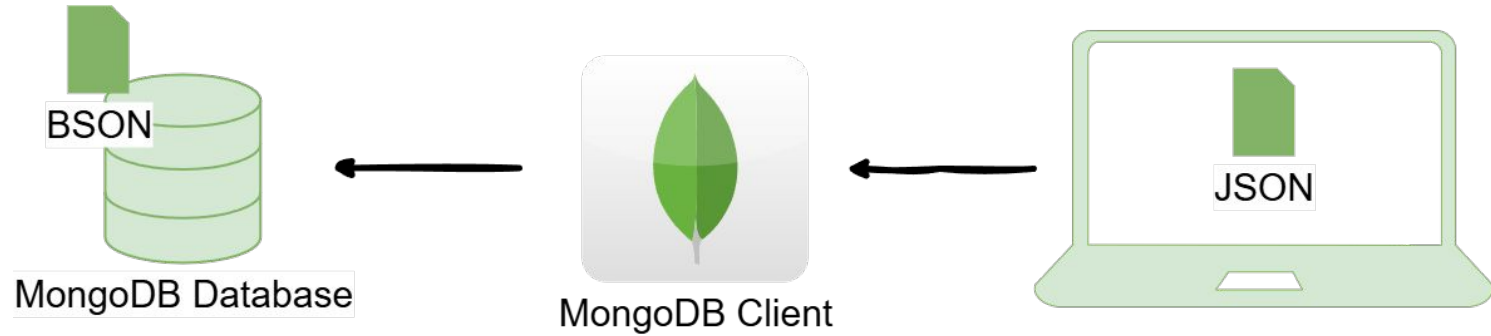


Advantages and Disadvantages

Advantages	Disadvantages
High Performance	Non-compliant to ACID
Horizontal Scalability	Dependent on application access
Dynamic and Flexible Schema	



MongoDB Backend Process



All documents come in JSON format. Once it's passed to the MongoDB Client through insert transactions, the client translates it to a BSON (Binary JSON) format and stores it into the database. This makes retrieval speeds much faster and makes the storage optimal.

A person with short dark hair, wearing a grey and black striped sweater, is seen from the back, looking at a wall covered in various design sketches, photos, and diagrams. A large white box with the word "Questions?" is superimposed over the person's head. The background is a collage of creative work, including wireframes, photos of people, and abstract diagrams. A dark blue arrow points from the left edge towards the top of the image.

Questions?

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Hands On Demo



Tasks

- Check MongoDB service status
- Connect to MongoDB
- List Databases, Collections
- Create Databases Rural Savings (ruralSavings) and Fitness Last (fitnessLast)
- Create Collections (ruralSavings.transactions, ruralSavings.accounts)
- Insert Document to each Collection
- Implicitly Create a Collection by inserting documents (fitnessLast.guestLogs)
- Show all records in each created collection



Check MongoDB Service Status

In your terminal, run the following commands:

```
sudo service mongod status
```

Or

```
sudo systemctl status mongod
```

– Ctrl + C to exit



Connect to MongoDB

In your terminal, run the following command:

```
mongosh
```



Create Databases

```
show databases;
```

```
show collections;
```

Note: Databases are implicitly created. It is created when a collection is created within that database.



Create Databases

```
use <databaseName>;
```

Note: Databases are implicitly created. It is created when a collection is created within that database.



Create Collections

```
db.createCollection("<collectionName>");
```

Note: Make sure that the database you intend to store your collection is selected before creating it. (use <databaseName>;)



InsertOne and InsertMany

```
db.<collectionName>.insertOne ({<object>} );
```

```
db.<collectionName>.insertMany ([<objects>] );
```

Note: Make sure that the database you intend to store your collection is selected before creating it. (use <databaseName>;)

End.

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